

Filter Design	
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Aim:

To verify the Filter Design.

Theoretical Calculation:

Circuit Diagram:

Ques Design a low pass filter using T section having a cut off freq of 2 kHz to operate with terminated load of 500 Ω ?

$$f_c = \frac{1}{\pi \sqrt{LC}} = 2 \text{ kHz}$$

$$= \frac{1}{\sqrt{LC}} = 2 \times \pi \times 1000 \text{ Hz}$$

$$= \left(\frac{1}{2\pi \times 1000} \right)^2 = LC$$

$$\Rightarrow 2.533 \times 10^{-8} = LC$$

$$K = \sqrt{\frac{L}{C}}$$

$$500 = \sqrt{\frac{L}{C}}$$

$$\frac{L}{C} = (500)^2$$

$$L = (500)^2 \times C$$

$$L = 250000 C$$

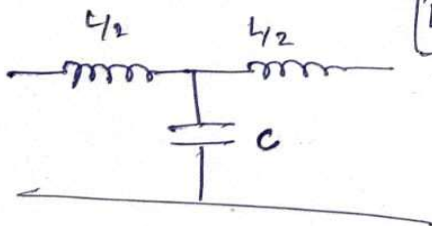
$$f_c = \frac{1}{\pi \sqrt{\frac{L}{C}} \sqrt{C}} = \frac{1}{\pi \sqrt{\frac{L}{C}} \times \sqrt{C}} = f_c \Rightarrow f_c = \left[\frac{1}{\pi K f_c} = C \right]$$

$$f_c = \frac{1}{\pi K \sqrt{C}} \Rightarrow \frac{1}{\pi \times 500 \sqrt{C}} = 2 \times 1000 \text{ Hz}$$

$$C = \left(\frac{1}{\pi \times 500 \times 2 \times 1000} \right)^2 = \left[3.1831 \times 10^{-7} \text{ F} = C \right]$$

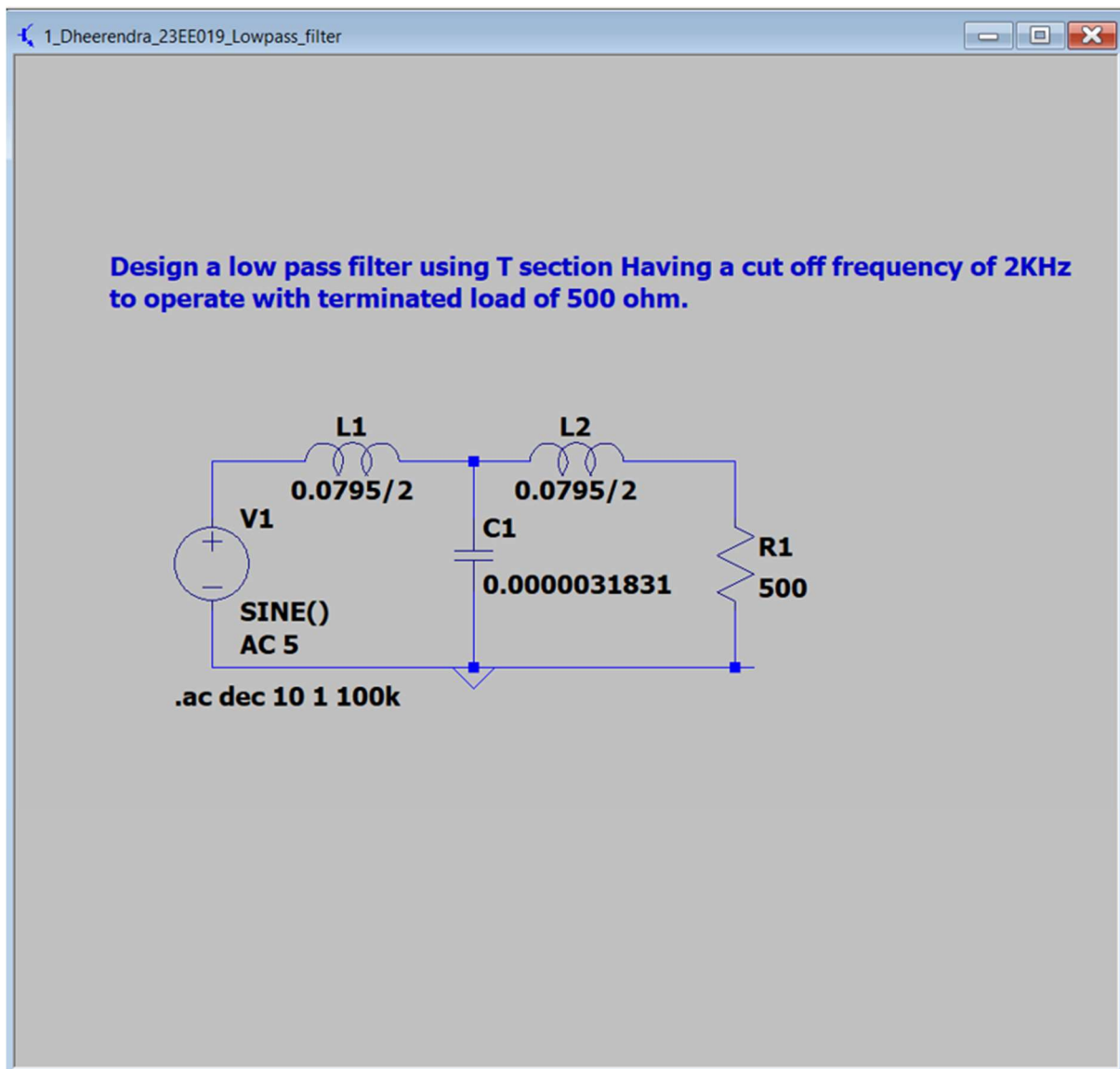
$$K^2 = \frac{L}{C} = (500)^2 \times 3.1831 \times 10^{-7} = L$$

$$L = 0.079 \text{ H}$$

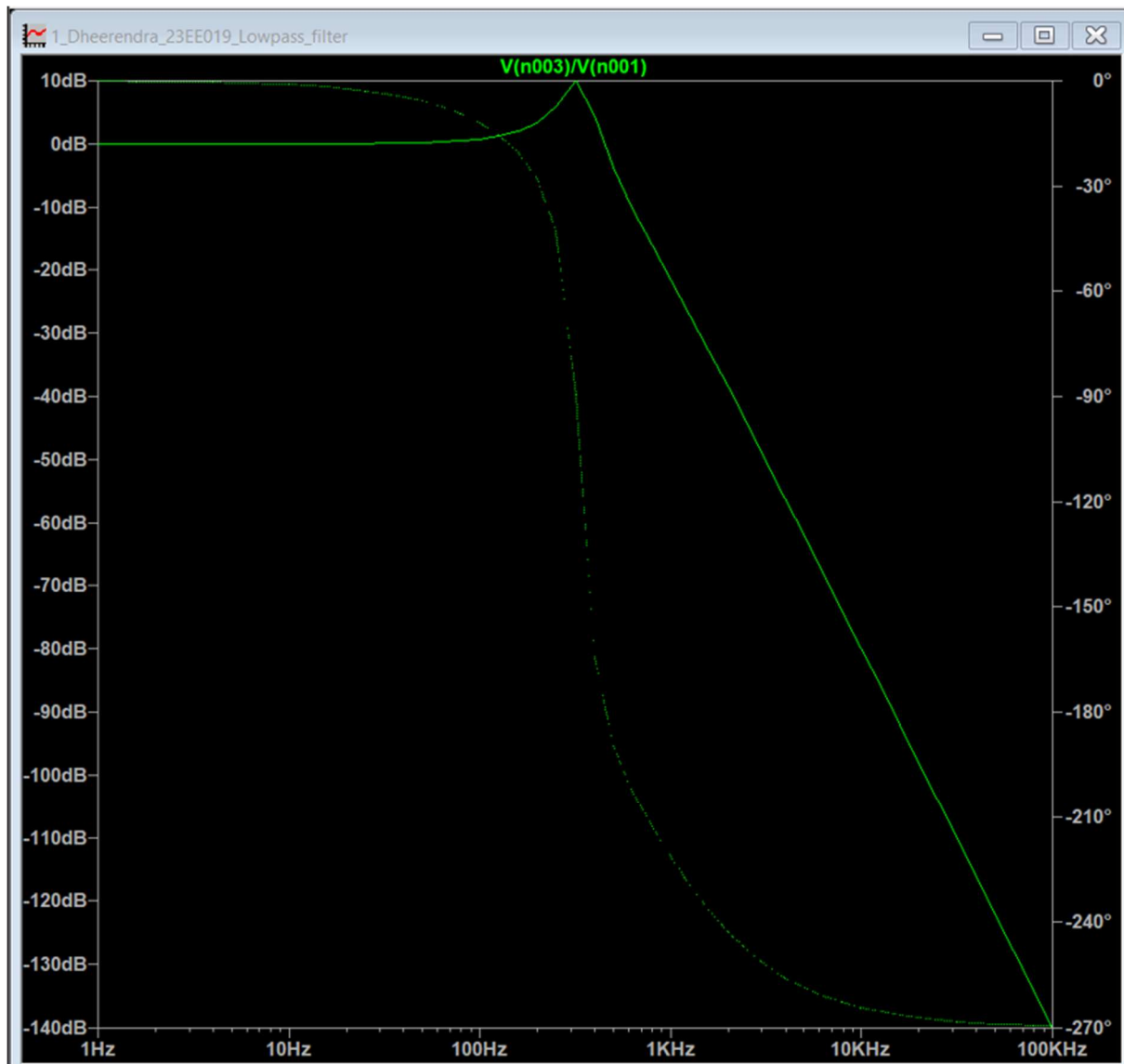


Verification by Simulation:

Circuit Diagram:



Plot of Frequency Response:



Results and Conclusion:

As we see that the output waveform of the evaluated circuit is the waveform of the Low pass filter which is given in the question. hence the filter circuit is verified.